TESTING REPORT

VisionRoot Service Request Management System

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1. Executive Summary

This report records verification activities completed for the VisionRoot Service Request Management System technical assignment. The latest end-to-end (E2E) rerun completed all seven test suites successfully, with **48 of 48 tests passing**. Frontend linting and production build, backend TypeScript compilation, and backend automated tests also completed successfully in the recorded runs.

2. Automated E2E Test Results

Test suite	Result	Outcome
Login	3/3	Passed
Registration	4/4	Passed
Protected routes	8/8	Passed
User workflow	8/8	Passed
Admin workflow	8/8	Passed
Admin status transitions	6/6	Passed
Admin cancellation	11/11	Passed
Total	**48/48**	**All passed**

The cancellation suite initially stopped because the two test request titles were not supplied to the separate Node.js process through environment variables. The titles printed by the setup script were assigned to VISIONROOT_PENDING_TITLE and VISIONROOT_INPROGRESS_TITLE in the PowerShell session. The rerun passed all 11 tests, including persistence verification after page refresh. This was a test invocation/configuration issue; no application source change was needed for the successful rerun.

3. Build and Static Analysis

Check	Recorded result
Frontend ESLint	Passed; no lint errors or warnings in the recorded run.
Frontend production build	Passed using TypeScript project build and Vite production build.
Backend TypeScript build	Passed; `tsc` completed without errors.
Backend automated tests	Passed: 6 test suites, 53 tests.

Build observation: Vite reported that the generated JavaScript bundle exceeded its 500 kB chunk-size advisory threshold (approximately 666.67 kB). The build still completed successfully. Code splitting and bundle optimization remain potential improvements.

4. Manual Verification Recorded

- User request workflow: create, search, view, edit, and verify updated request information.
- Admin dashboard: statistics, recent requests, search, status filtering, and navigation.

- Admin status handling: status changes and terminal-state behavior were checked through the UI.
- Admin user management: user listing, search, role changes, and safeguards against self-demotion or demoting the last administrator.
- Support Centre: customer/admin replies, status progression, closing a ticket, and rejection of replies to a closed ticket.
- Health endpoint: HTTP 200 response was observed.

5. Test Environment

- Operating system: Windows 11 (24H2).
- Frontend: Vite development server at http://localhost:5173; backend: Express API at http://localhost:5000.
- Database: MongoDB Atlas.
- Browser automation: Selenium WebDriver with Chrome.
- Backend test runner: Jest (as configured in the project).

6. Notes and Remaining Considerations

- The 48/48 E2E total represents the seven suites rerun and shown in the recorded terminal outputs.
- Backend automated test result is 53/53 across six suites from the recorded run.
- Chrome updater access-denied messages appeared during the cancellation run, but the suite completed with 11/11 passing. These messages did not cause test failures.
- A TensorFlow Lite XNNPACK delegate informational message appeared during browser execution; the test suite still passed.
- Responsive layouts should receive a final direct visual review at representative desktop and mobile viewport sizes if time permits.
- The frontend bundle-size advisory is non-blocking and should be considered for future optimization.

7. Conclusion

Based on the recorded checks, the application passed the latest 48 E2E tests, the backend 53-test automated suite, frontend linting, and the frontend/backend build checks. The tested user and administrator workflows behaved as expected in these runs. The observations listed above are retained as follow-up considerations rather than reported as test failures.